

Human Protein Atlas Image Classification Research

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1 Problem definition

The Human Protein Atlas Image Classification project aims to enhance understanding and predictions of protein subcellular localization using multi-label classification methods. Protein localization within cells is vital to unraveling biological processes, understanding disease mechanisms, and identifying potential therapeutic targets. This Kaggle competition dataset provides fluorescence microscopy images where each image comprises multiple color channels representing distinct cellular structures, including the nucleus, microtubules, and the endoplasmic reticulum, along with the protein of interest.

Participants are tasked with accurately classifying the subcellular localization patterns of proteins from these complex immunofluorescence images. The multi-label nature of this challenge means that proteins may localize simultaneously in multiple cellular compartments, requiring advanced modeling strategies capable of identifying multiple concurrent labels. Achieving high accuracy in this task has significant implications, potentially accelerating research in areas such as drug discovery, disease diagnosis, and personalized medicine.

The complexity of protein patterns, variable staining intensities, overlapping cellular structures, and inherent image noise pose significant hurdles. To overcome these, advanced image preprocessing techniques such as intensity normalization, contrast enhancement, and noise reduction are often necessary. Additionally, sophisticated deep learning methods, such as transformer-based architectures and specialized loss functions, are explored to handle the nuances and variability in the data.

1.1 Dataset Overview

The Human Protein Atlas (HPA) Image Classification dataset provides high-resolution microscopy images for classifying subcellular protein localization patterns. It was created using standardized

confocal microscopy techniques and is widely used for studying protein functions and cellular structures.

The dataset contains 31,072 samples, each represented by four grayscale images corresponding to different fluorescence channels: nucleus (blue), microtubules (red), endoplasmic reticulum (yellow), and the protein of interest (green). The green channel is the primary signal for classification, while the other channels provide structural context. The images are available in 512x512 resolution for computational efficiency and high-resolution formats for more detailed analysis.

Each sample is labeled with one or more of 28 localization categories, making this a multi-label classification task. The dataset covers 27 different cell types, each with distinct morphology, influencing the observed protein patterns. A major challenge is class imbalance, requiring methods like weighted loss functions or augmentation to improve model performance.

This dataset serves as a benchmark for deep learning applications in biomedical imaging, particularly in classifying complex protein distributions. It supports the development of automated methods for analyzing cellular structures, with potential applications in computational biology and medical research.

2 Key references

According to the leaderboard, the best solution was provided by the bestfitting team (Patil, 2019), who used a ResNet50 backbone followed by a classifier head trained with ArcFaceLoss (Deng et al., 2022). Although that approach was published in 2019 using ResNet50, newer architectures such as Swin Transformer (Liu et al., 2021) and Vision Transformer (ViT) (Dosovitskiy et al., 2021) have since emerged, raising the question of whether replacing ResNet50 with these models could yield better performance.

Similarly, the solution’s classifier head was trained with ArcFaceLoss, first introduced around 2018–2019. Since then, other methods like CurricularFace (Huang et al., 2020) (published in 2020) and AdaFace (Kim et al., 2023) (published in 2023) have been proposed. We therefore wonder whether employing these newer loss functions during training might further enhance performance.

3 Relationship to team members’ backgrounds

Our team has experience in machine learning and deep learning, particularly in unsupervised learning, feature extraction, and embedding-based modeling. Additionally, we have practical experience in image processing and editing, including tasks such as image enhancement, denoising, segmentation, and visualization. Some members have worked on emerging keyword extraction, paper section title normalization, and lyric popularity prediction, which involved designing algorithms to uncover patterns in data without explicit labels.

This project presents an opportunity to apply our expertise in representation learning, optimization strategies, image processing, and model evaluation to the domain of computer vision, exploring how different backbones (ResNet vs. Swin Transformer) and loss functions (ArcFace vs. AdaFace) impact performance. By drawing on our prior experience with unsupervised learning, image editing, and deep learning model comparisons, we aim to conduct a rigorous analysis of state-of-the-art classification techniques.

References

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